NO.7 LIMITED i-stop FUNCTION OPERATION TIME [SKYACTIV-D 1.5]

id1103q4001200

7	LIMITED i-stop FUNCTION OPERATION TIME
DESCRIPTION	• Frequency of which i-stop function operates is low. • Frequent occurrence of engine restarting other than for driving vehicle from stop.
POSSIBLE CAUSE	**Note** • To ensure electric vehicle system reliability, restart the engine regardless of the vehicle condition after **120 s** have elapsed since the i-stop function operated. **Battery voltage low during i-stop function operation** • Battery deterioration • Insufficient battery recharge while engine is running — Low recharge effect • Generator malfunction (part, system, control malfunction) • Large amount of vehicle power consumption • High electrical load from aftermarket electrical accessories **False detection of engine starting conditions** • False detection of battery voltage • False detection of low power brake unit load — Power brake unit vacuum sensor malfunction — Short or open circuit in wiring harness between the following terminals: • Power brake unit vacuum sensor terminal A—PCM terminal 2AK • Power brake unit vacuum sensor terminal B—PCM terminal 2BD • Power brake unit vacuum sensor terminal C—PCM terminal 2BC — Power brake unit malfunction (air tightness malfunction) — Malfunction in vacuum hose to power brake unit (damage, bad check valve) • Climate control unit falsely recognizes MAX HOT or MAX COLD of air mix door on driver-side (with full-auto air conditioner) — Driver-side air mix actuator malfunction — Driver-side air mix actuator position sensor malfunction — Driver-side air mix door link stuck **False recognition of driver performing engine start operation** • False detection of depressed clutch pedal (MTX) — Clutch stroke sensor malfunction — Short to power supply in wiring harness between clutch stroke sensor terminal B and PCM terminal 2AJ • False detection of released brake pedal (ATX) — Brake fluid pressure sensor (built-into DSC HU/CM) malfunction • False detection of steering wheel operation (ATX) — Steering angle sensor initialization malfunction — EPS control module malfunction

Diagnostic Procedure

STEP	INSPECTION	RESULTS	ACTION
1	**INSPECT EFFECT OF NON-GENUINE ELECTRICAL ACCESSORY FOR CAUSE OF MALFUNCTION** • Remove any non-genuine electrical accessory. • Verify the malfunction symptom. • Is the frequency of the engine restarting from the i-stop off condition the same as that of another vehicle of the same model?	Yes	The system is normal. • Explain to the customer that the frequency of the engine restarting increases due to the effect of the non-genuine electrical accessory installed.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
2	**VERIFY DTC** • Retrieve the PCM, TCM, body control module (BCM), DSC HU/CM, instrument cluster and climate control unit DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].) (See DTC INSPECTION [BCM].) (See DTC INSPECTION [DSC HU/CM].) (See DTC INSPECTION [INSTRUMENT CLUSTER].) (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].) (See DTC TABLE [BCM].) (See DTC TABLE [DSC HU/CM].) (See DTC TABLE [INSTRUMENT CLUSTER].) (See DTC TABLE [FULL-AUTO AIR CONDITIONER].)
		No	Go to the next step.
3	**VERIFY BATTERY CONDITION** • Inspect the battery. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Go to the next step.
		No	Go to Step 5.
4	**DETERMINE IF MALFUNCTION CAUSE IS BATTERY OR GENERATOR** • Recharge the battery. (See BATTERY RECHARGING.) • Inspect the battery again. (See BATTERY INSPECTION.) • Is there any malfunction?	Yes	Replace the battery. (See BATTERY RECHARGING.)
		No	Go to the next step.
5	**INSPECT GENERATOR** • Inspect the generator. (See GENERATOR INSPECTION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR INSPECTION [WITH i-ELOOP (SKYACTIV-D 1.5)].) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results.
		No	ATX: • Go to the next step. MTX: • Go to Step 8.
6	**DETERMINE IF MALFUNCTION IS CAUSED BY STEERING ANGLE (ESTIMATED ABSOLUTE ANGLE) SIGNAL ERROR** • Start the engine and idle it. • Access the EPS control module PID STR_AB_EST using the M-MDS. (See PID/DATA MONITOR INSPECTION [EPS CONTROL MODULE].) • Is the PID value normal?	Yes	Go to Step 8.
		No	Go to the next step.
7	**INSPECT EPS CONTROL MODULE FOR MALFUNCTION** • Inspect the EPS control module. (See EPS CONTROL MODULE INSPECTION.) • Is the EPS control module normal?	Yes	Perform the following procedure: 1. Switch the ignition to OFF, and maintain this condition for approx. 3 s. 2. Set the vehicle wheels in the straight-ahead position. 3. Switch the ignition ON (engine off or on). 4. Using the M-MDS, display EPS control module PID STR_AB_EST. (See PID/DATA MONITOR INSPECTION [EPS CONTROL MODULE].) • If the STR_AB_EST value is normal, go to Step 17. (Because the steering angle (estimated absolute angle) has returned to normal) • If the STR_AB_EST value is not normal, replace the EPS control module, then go to Step 17. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
		No	Replace the EPS control module, then go to Step 17. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)

STEP	INSPECTION	RESULTS	ACTION
8	**DETERMINE IF MALFUNCTION CAUSE IS POWER BRAKE UNIT VACUUM SENSOR SIGNAL OR OTHER** • Put the vehicle in an i-stop condition (engine stopped). • Access the PCM PID BBP using the M-MDS with the brake pedal depressed and the engine stopped. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Does the BBP PID value remain **less than −45 kPa {−0.46 kgf/cm^2, −6.5 psi}**?	Yes	Vehicles with manual air conditioner: • ATX: — Go to Step 14. • MTX: — Go to Step 15. Vehicles with full-auto air conditioner: • Go to Step 12.
		No	Go to the next step.
9	**INSPECT POWER BRAKE UNIT VACUUM SENSOR FOR AIR TIGHTNESS MALFUNCTION** • Perform the vacuum function inspection for the power brake unit and the vacuum loss inspection. (See POWER BRAKE UNIT INSPECTION.) • Is there any malfunction?	Yes	Repair or replace the malfunctioning part according to the inspection results. Go to Step 17.
		No	Go to the next step.
10	**INSPECT POWER BRAKE UNIT VACUUM SENSOR** • Inspect the power brake unit vacuum sensor. (See POWER BRAKE UNIT VACUUM SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the power brake unit vacuum sensor. (See POWER BRAKE UNIT REMOVAL/INSTALLATION [L.H.D.].) (See POWER BRAKE UNIT REMOVAL/INSTALLATION [R.H.D. (SKYACTIV-D 1.5)].)
		No	Go to the next step.
11	**INSPECT POWER BRAKE UNIT VACUUM SENSOR CIRCUIT FOR SHORT TO GROUND OR OPEN CIRCUIT** • Inspect for open or short circuit between the following terminals (wiring harness-side): — Power brake unit vacuum sensor terminal A—PCM terminal 2AK — Power brake unit vacuum sensor terminal B—PCM terminal 2BD — Power brake unit vacuum sensor terminal C—PCM terminal 2BC • Is there any malfunction?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Power brake unit vacuum sensor terminal A—PCM terminal 2AK • Power brake unit vacuum sensor terminal B—PCM terminal 2BD • Power brake unit vacuum sensor terminal C—PCM terminal 2BC **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground or open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground or open circuit. Go to step 17.
		No	Replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
12	**DETERMINE IF MALFUNCTION CAUSE IS DRIVER-SIDE AIR MIX ACTUATOR SIGNAL OR OTHER** • Measure the voltage at the following terminal (wiring harness-side) when the driver-side temperature setting is MAX HOT and MAX COLD. — Climate control unit terminal 2E • Is the voltage normal? (See CLIMATE CONTROL UNIT INSPECTION [FULL-AUTO AIR CONDITIONER].)	Yes	ATX: • Go to Step 14. MTX: • Go to Step 15.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
13	**INSPECT DRIVER-SIDE AIR MIX ACTUATOR** • Inspect the driver-side air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is there any malfunction?	Yes	Replace the driver-side air mix actuator. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) Go to step17.
		No	Inspect the air mix actuator and linkage for sticking. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results. Go to step 17.
14	**DETERMINE IF MALFUNCTION CAUSE IS BRAKE FLUID PRESSURE SENSOR SIGNAL OR OTHER** • Put the vehicle in an i-stop condition (engine stopped). • Monitor the PCM PID BFP using the M-MDS while the brake is depressed and held with the i-stop function operating. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Does the monitoring value change?	Yes	Repeat the inspection from Step 1. • If the malfunction is not resolved, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 17.
		No	Brake fluid pressure sensor malfunction. • Replace the DSC HU/CM. (See DSC HU/CM REMOVAL/INSTALLATION [L.H.D.].) (See DSC HU/CM REMOVAL/INSTALLATION [R.H.D.].)
15*	**DETERMINE IF MALFUNCTION CAUSE IS CLUTCH STROKE SENSOR SIGNAL OR OTHER** • Switch the ignition ON (engine off). • Access the PCM PID CPP using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Does the CPP PID value change according to the amount the clutch pedal is depressed?	Yes	Repeat the inspection from Step 1. • If the malfunction is not resolved, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 17.
		No	Go to the next step.
16	**INSPECT CLUTCH STROKE SENSOR** • Inspect the clutch stroke sensor. (See CLUTCH STROKE SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Replace the clutch master cylinder. (See CLUTCH MASTER CYLINDER REMOVAL/INSTALLATION [F66M-R].)
		No	Inspect for short to power supply between clutch stroke sensor terminal B and PCM terminal 2AJ. • If there is any malfunction: — Refer to the wiring diagram and verify whether or not there is a common connector between clutch stroke sensor terminal B and PCM terminal 2AJ. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. • If there is no malfunction: — Replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
17	Verify the test results. • If normal, return to the diagnostic index to service any additional symptoms. (See SYMPTOM DIAGNOSTIC INDEX [SKYACTIV-D 1.5].) • If a malfunction remains, inspect the related Service Information and perform the repair or diagnosis. — If the vehicle is repaired, troubleshooting is completed. — If the vehicle is not repaired or additional diagnostic information is not available, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)		